Team Name : Impacteers

Team Leader Name: Likith Talla

Problem Statement : Generative AI-Based Cloud Removal and Reconstruction for LISS-IV Satellite Imagery

Team Members

Team Leader
Name : Likith Talla
College : Swami Vivekananda Institute of Technology

Team Member - 1
Name : Ananya Pappu
College : Swami Vivekananda Institute of Technology

Team Member - 2
Name : Aman Kumar Sharma
College : Swami Vivekananda Institute of Technology

Team Member - 3
Name : A.J.Abhishek
College : Swami Vivekananda Institute of Technology

How Our Idea Differs from Existing Solutions

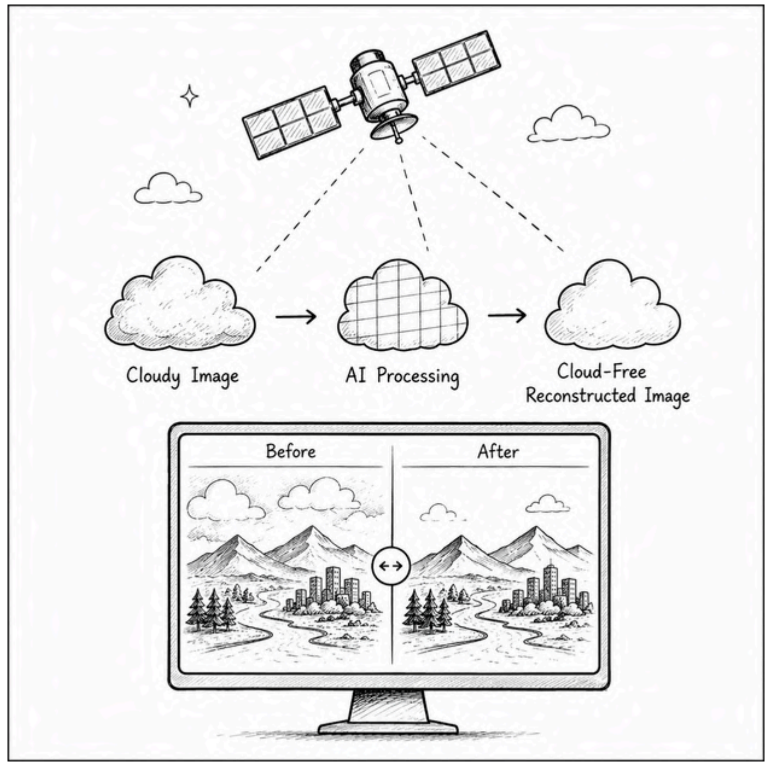
- **End-to-End AI Pipeline** integrating **Cloud Detection, Cloud Segmentation, Generative Reconstruction,** and **Interactive Image Comparison** instead of standalone modules.
- Optimized for **LISS-IV** with support for **Sentinel-2, Landsat,** and **Sentinel-1 SAR**, unlike existing solutions focused mainly on Sentinel-2/Landsat.
- Replaces **Cloud Masking, Image Inpainting,** and **Multi-Temporal Image Replacement** with **U-Net** and **Pix2Pix GAN** for **AI-driven Surface Reconstruction**.
- Provides **Real-Time Reconstruction Workflow** and **Before–After Visualization**, unlike traditional **algorithm-centric** research models.
- Built as a **Scalable, User-Centric Earth Observation Platform** for practical **Remote Sensing** applications.

How Our Solution Solves the Problem

- Detects cloud regions using a **U-Net-based Cloud Segmentation Model.**
- Reconstructs hidden land features using **Generative AI (Pix2Pix GAN).**
- Leverages **Multi-Temporal** and **Multi-Sensor Satellite Data (LISS-IV, Sentinel-2, Landsat)** to improve reconstruction quality.
- Produces **High-Fidelity Cloud-Free Satellite Imagery** with minimal information loss.
- Delivers results through an **End-to-End Automated Pipeline** for **Remote Sensing and Earth Observation** applications.

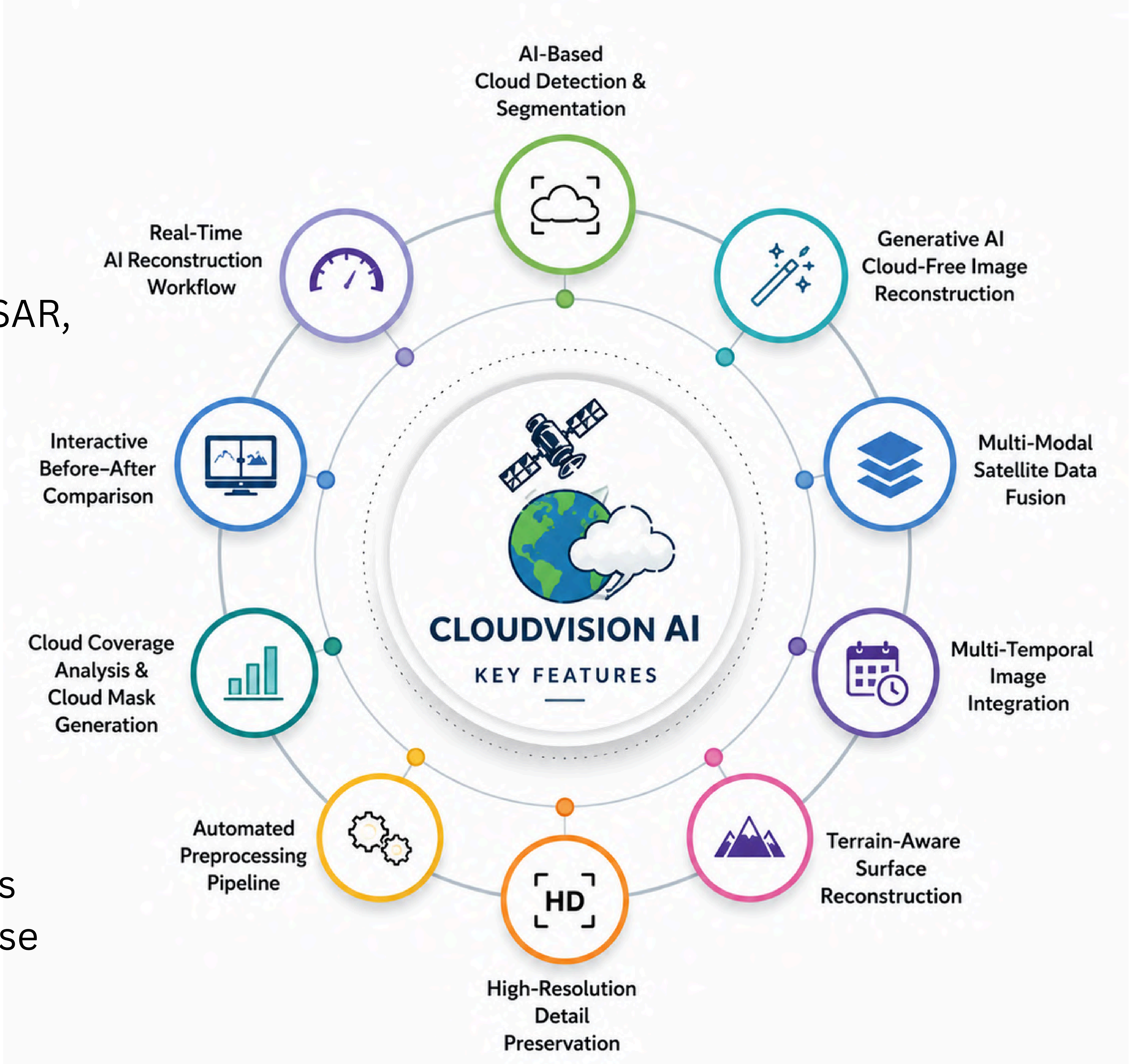
USP of the Proposed System

- **AI-Driven Reconstruction** using **U-Net** and **Pix2Pix GAN** for realistic cloud-free image generation.
- **Multi-Modal Satellite Fusion** integrating **LISS-IV, Sentinel-1 SAR, Sentinel-2, Landsat,** and **Temporal Data.**
- **Terrain-Aware Reconstruction** preserving region-specific features such as **Urban, Agriculture, Forest, and Water Bodies**.
- **High-Resolution Preservation** with enhanced **Spatial Detail** and **Surface Texture Reconstruction.**
- **Scalable Architecture** designed for National **Geospatial Datasets** and large-scale **Earth Observation** applications.
- **Supports Disaster Response, Precision Agriculture, Environmental Monitoring,** and **Geospatial Decision Support.**



Features Offered

✔ AI-Based Cloud Detection & Segmentation
✔ Generative AI Cloud-Free Image Reconstruction
✔ Multi-Modal Satellite Data Fusion (LISS-IV, Sentinel-1 SAR, Sentinel-2, Landsat)
✔ Multi-Temporal Image Integration
✔ Terrain-Aware Surface Reconstruction
✔ High-Resolution Spatial Detail Preservation
✔ Automated Image Preprocessing Pipeline
✔ Cloud Coverage Analysis & Cloud Mask Generation
✔ Real-Time AI Reconstruction Workflow
✔ Interactive Before–After Image Comparison
✔ Image Quality Evaluation (SSIM, PSNR, RMSE)
✔ Downloadable Cloud-Free Satellite Images
✔ Scalable Architecture for National Geospatial Datasets
✔ Applications in Precision Agriculture, Disaster Response & Environmental Monitoring



PROCESS FLOW DIAGRAM

1 START
User initiates the process

2 INPUT SATELLITE IMAGES (6 DATASETS)
- LISS IV (Target Dataset)
- Sentinel-1 SAR
- Sentinel-2
- Landsat
- DEM Data
- Temporal Imagery

3 REGION WISE SELECTION
Select target region
- Mizoram
- Guwahati
- Shillong
- Itanagar

4 IMAGE PRE-PROCESSING (5 STEPS)
- Radiometric Correction
- Atmospheric Correction
- Geometric Correction
- Image Normalization
- Resizing / Resampling
- Band Selection

5 CLOUD DETECTION (3 STEPS)
Using U-Net Model
- Cloud Detection
- Cloud Segmentation
- Cloud Mask Generation

6 CLOUD MASK ANALYSIS
Analyze cloud mask to extract information
- Cloud Coverage %
- Cloud Location
- Occluded Regions

7 GENERATIVE AI RECONSTRUCTION
Using Pix2Pix GAN
- Reconstruct Cloud Covered Regions
- Surface Reconstruction
- Texture Generation

8 IMAGE ENHANCEMENT
Improve visual quality
- Contrast Enhancement
- Sharpness Improvement
- Noise Reduction
- Color Balancing

9 QUALITY EVALUATION (3 METRICS)
Evaluate reconstructed image quality
- SSIM (Structural Similarity)
- PSNR (Peak Signal to Noise)
- RMSE (Root Mean Square Error)

10 CLOUD-FREE IMAGE GENERATION
Generate final cloud-free image

11 COMPARISON
Compare before and after results
- Before vs After View
- Difference Visualization
- Improvement Analysis

12 DOWNLOAD
Download the reconstructed cloud-free image

OUTPUT
High Quality Cloud-Free Reconstructed Image

APPLICATIONS
Precision Agriculture
Disaster Management
Environmental Monitoring
Urban Planning
Forest Monitoring
Water Resource Management

TECHNOLOGIES USED
U-Net
Pix2Pix GAN
Python
OpenCV
TensorFlow / PyTorch
GIS Integration

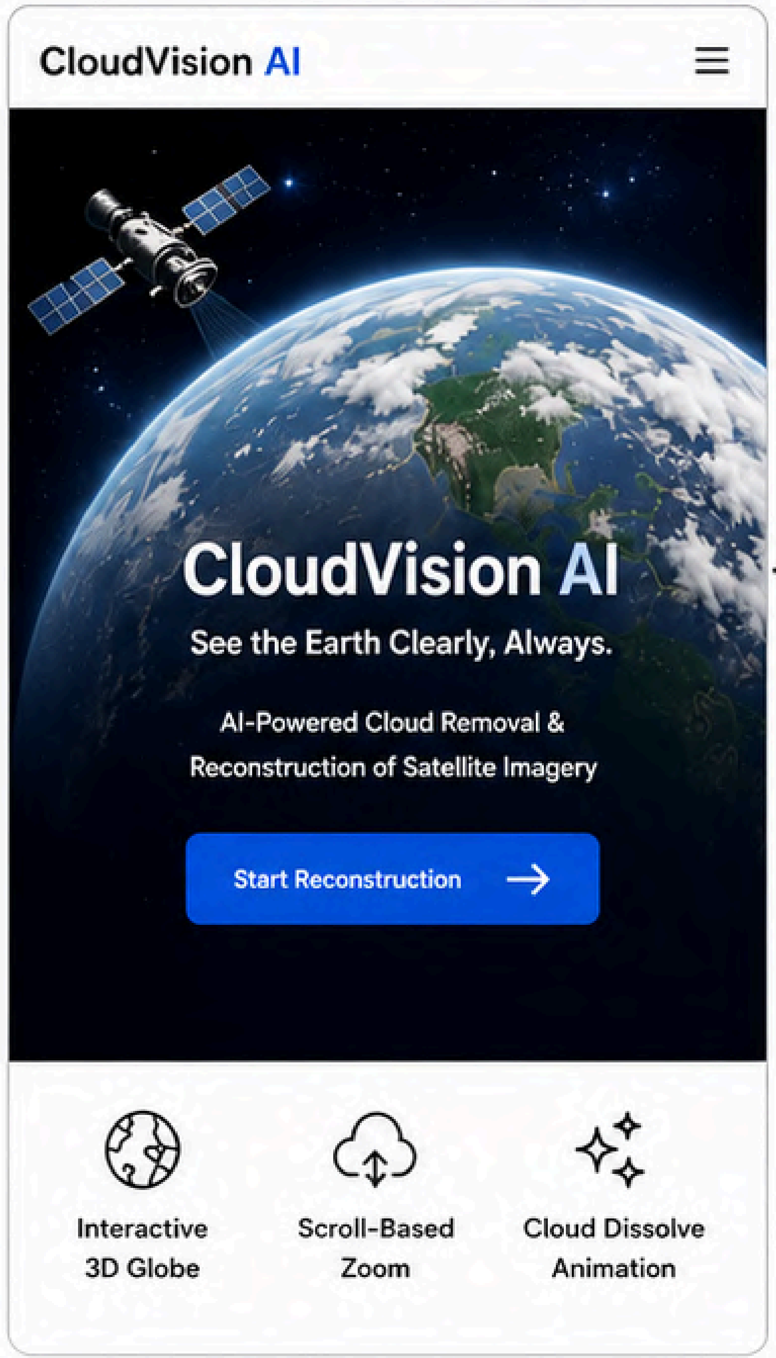
Input Data → Pre-processing → Cloud Detection → AI Reconstruction → Enhancement → Evaluation → Cloud-Free Image → Comparison → Download

WIREFRAMES / MOCK DIAGRAMS OF THE PROPOSED SOLUTION

CloudVision AI – AI-Powered Cloud Removal & Reconstruction Platform

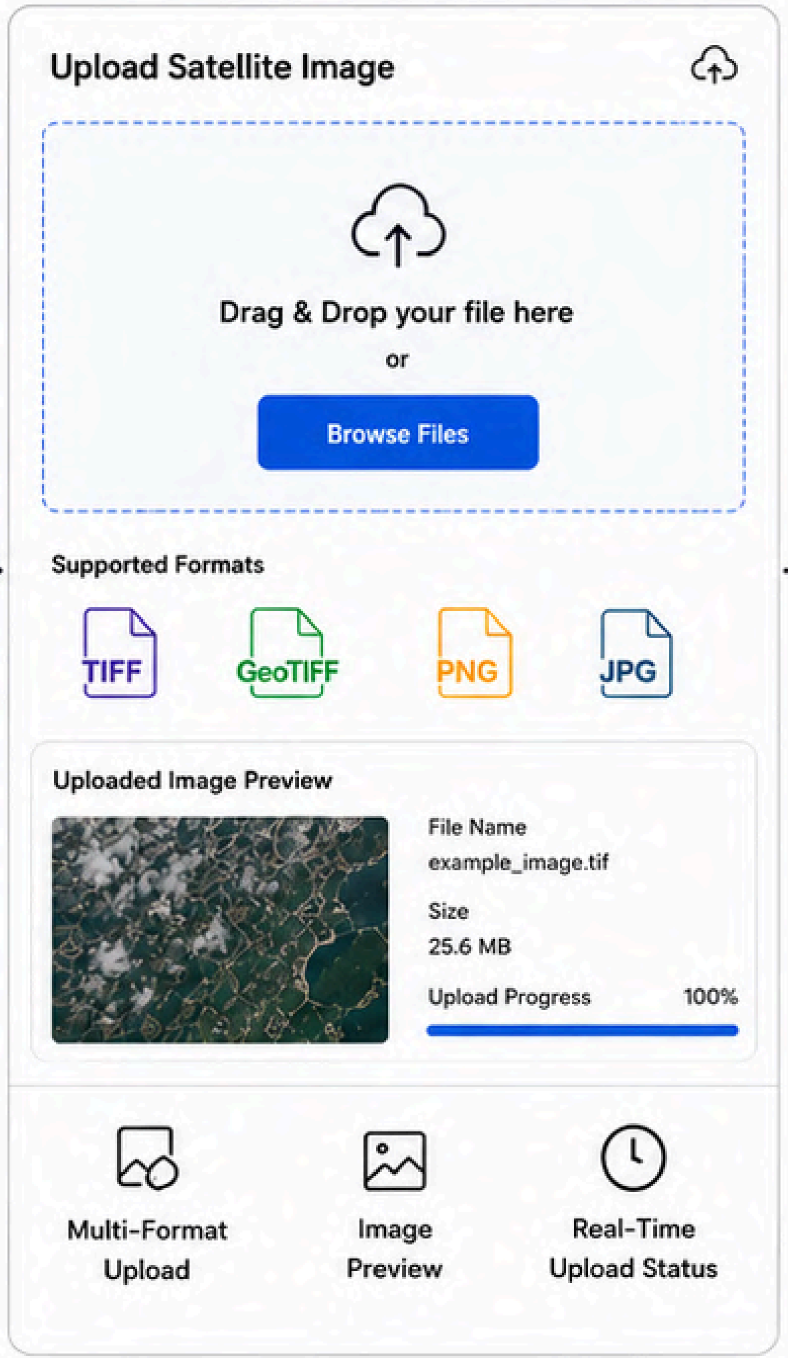
1 HOME PAGE

Introduce the platform



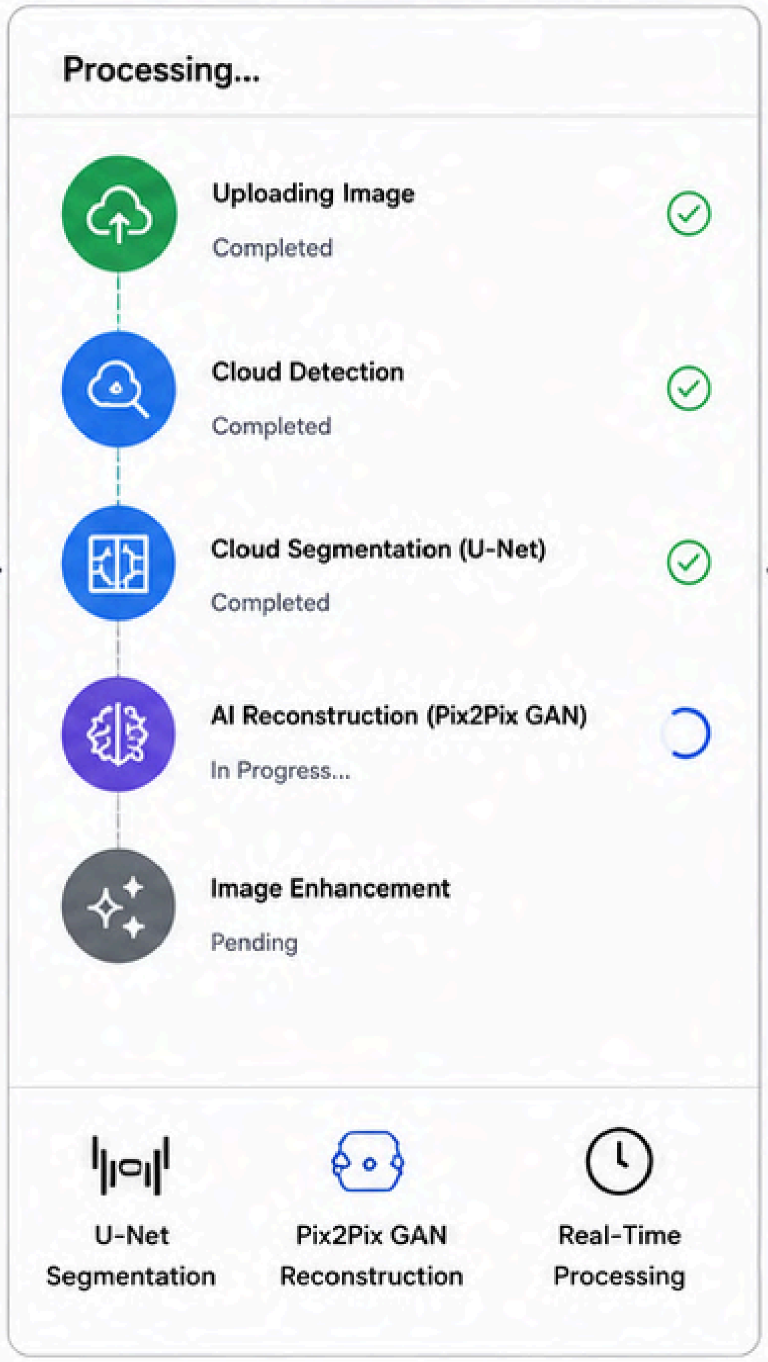
2 IMAGE UPLOAD

Upload satellite imagery



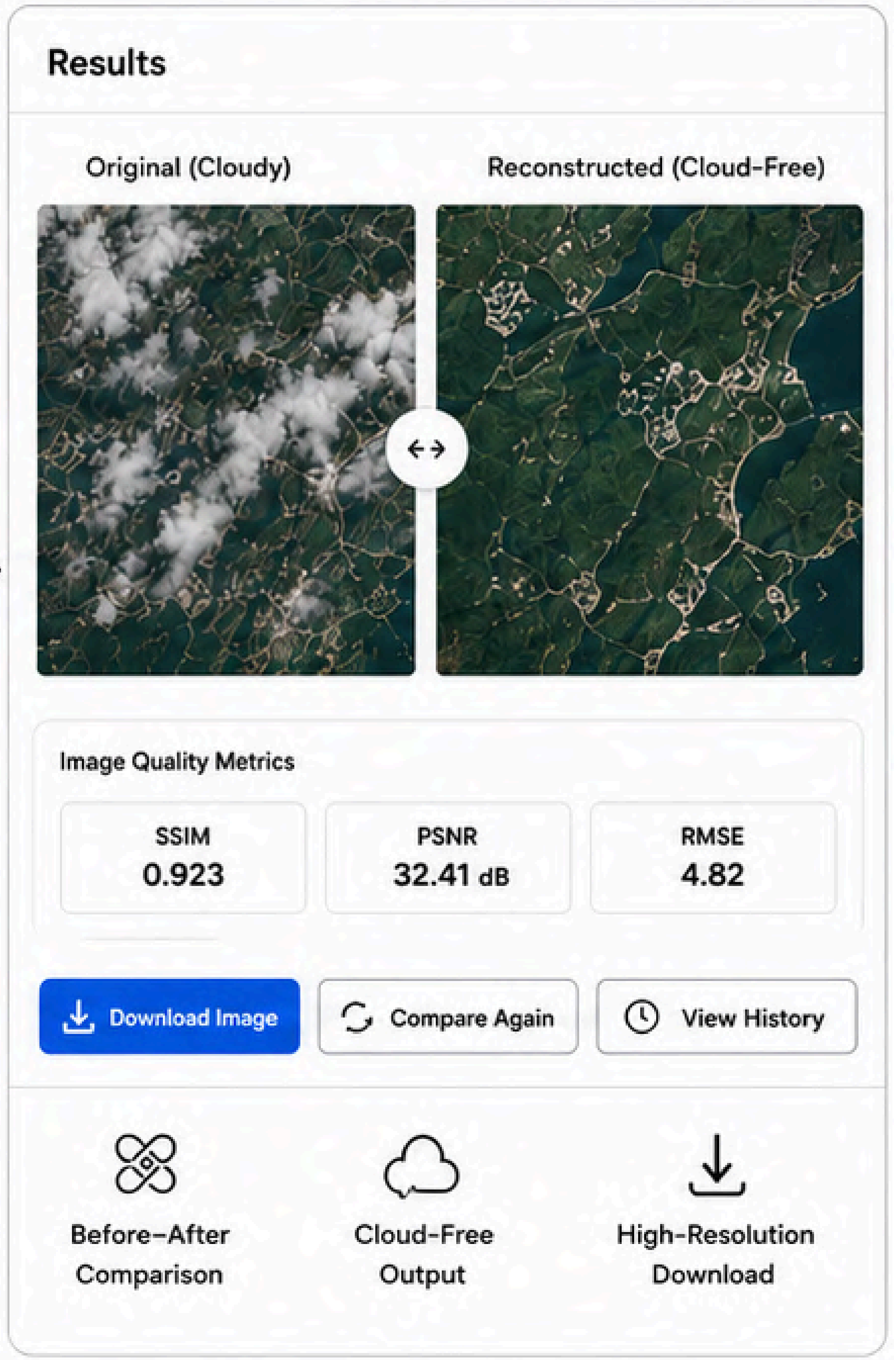
3 AI RECONSTRUCTION

AI is processing the image



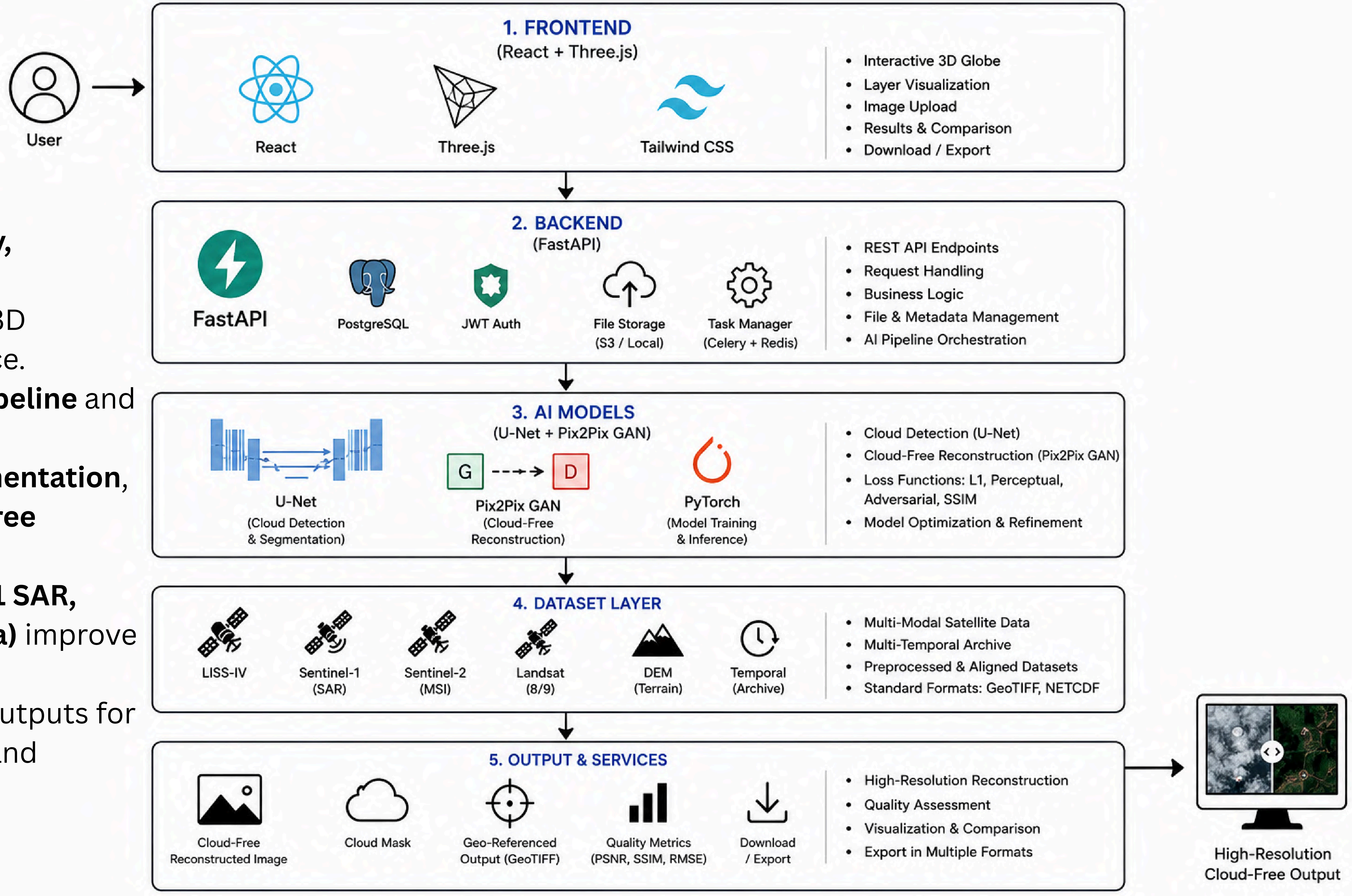
4 RESULTS & COMPARISON

View & compare results



Architecture diagram

- Layered Architecture enables **Modularity, Scalability,** and **easy maintenance**.
- **React + Three.js** provides an interactive 3D Visualization and intuitive user experience.
- **FastAPI** orchestrates the **AI Inference Pipeline** and **manages backend** services efficiently.
- **U-Net** performs **Cloud Detection & Segmentation**, while **Pix2Pix GAN reconstructs Cloud-Free Satellite Imagery.**
- Multi-Modal Datasets **(LISS-IV, Sentinel-1 SAR, Sentinel-2, Landsat, DEM, Temporal Data)** improve reconstruction accuracy.
- Generates High-Resolution Cloud-Free Outputs for **Remote Sensing, Precision Agriculture,** and **Disaster Management**



TECHNOLOGIES TO BE USED IN THE PROPOSED SOLUTION

CATEGORY	TECHNOLOGIES	PURPOSE
FRONTEND	React.js, TypeScript, Tailwind CSS, Three.js, React Three Fiber	Build interactive and responsive 3D web interface for visualization and user interaction
BACKEND	FastAPI, Python	Develop RESTful APIs, handle requests, and integrate AI models
AI / DEEP LEARNING	PyTorch, U-Net, Pix2Pix GAN	Cloud detection (U-Net) and cloud-free image reconstruction (Pix2Pix GAN)
IMAGE PROCESSING	OpenCV, NumPy, Pillow	Image preprocessing, manipulation, and enhancement for better model performance
GEOSPATIAL DATA	GDAL, Rasterio, GeoPandas	Handle geospatial data, read/write GeoTIFF, and perform spatial transformations
DATASETS	LISS-IV, Sentinel-1 SAR, Sentinel-2, Landsat, DEM, Temporal Imagery	Multi-modal and multi-temporal satellite data for training and reconstruction
MODEL EVALUATION	SSIM, PSNR, RMSE	Evaluate reconstruction quality using standard metrics
DEPLOYMENT	Docker, GitHub, Vercel/Render	Containerization, version control and cloud deployment

BHARATIYA ANTARIKSH HACKATHON 2026

THANK YOU

Team Impacteers